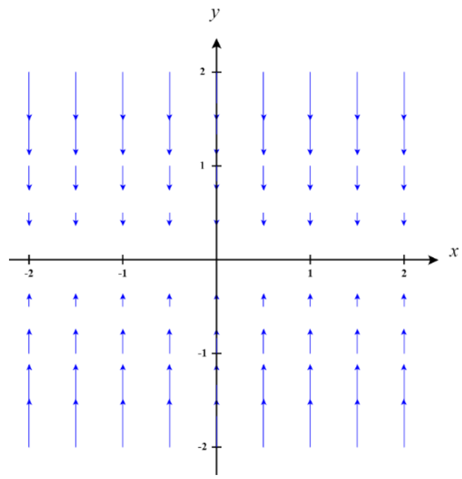
MTH 202 Chapter 5: Topics In Vector Calculus

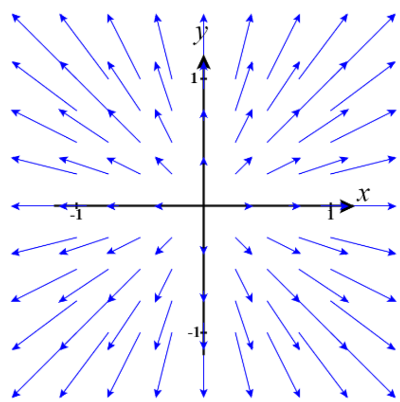
5.1 Vector Field Worksheet

1. Sketch the vector field $\mathbf{F}(x,y) = -y\mathbf{j}$.

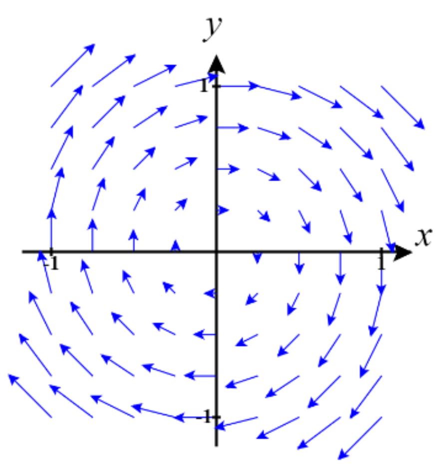


2. Let $f(x,y) = x^2 + y^2$. Sketch the vector field $\nabla f(x,y)$

   First, we calculate that $\nabla f(x,y) = \langle 2x, 2y \rangle$.



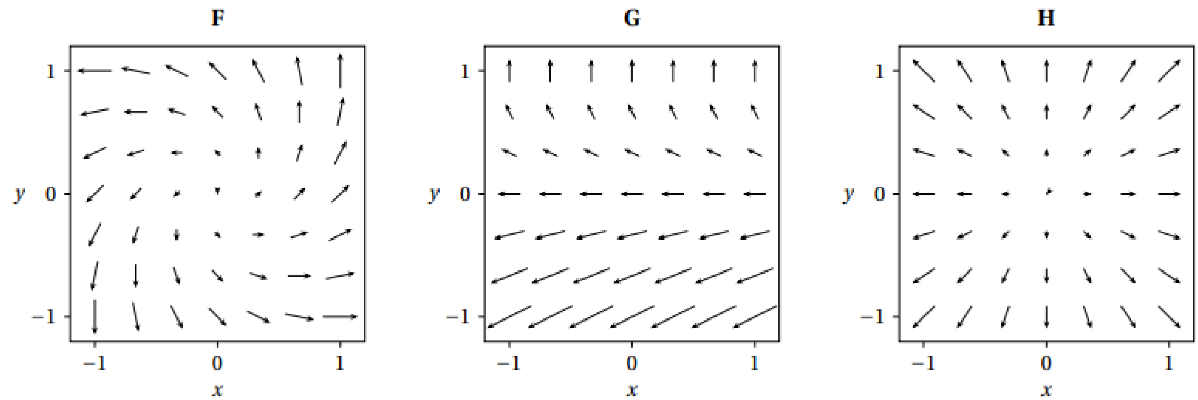
3. Sketch the vector field $\mathbf{F}(x,y) = \langle y, -x \rangle$.



4. Compute the gradient vector field of *f*, where $f = \arctan\left(\frac{x}{y}\right)$

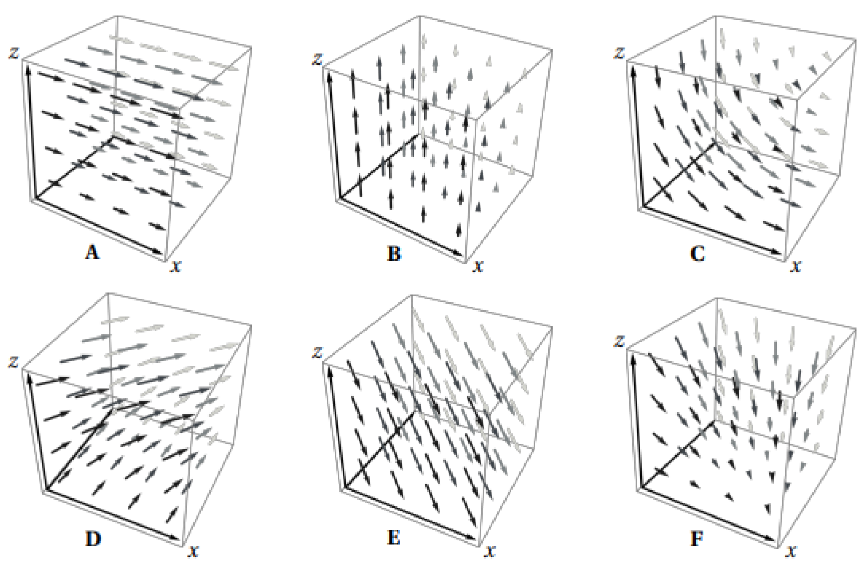
First, we calculate that $\nabla f(x,y) = \left\langle \frac{1}{y\left(1+x^2/y^2\right)}, \frac{-x}{y^2\left(1+x^2/y^2\right)} \right\rangle = \left\langle \frac{1}{y+x^2/y}, \frac{-x}{y^2+x^2} \right\rangle$.

5. Three vector fields are shown below. Which of the vector field corresponds to $\langle y-1, y\rangle$?



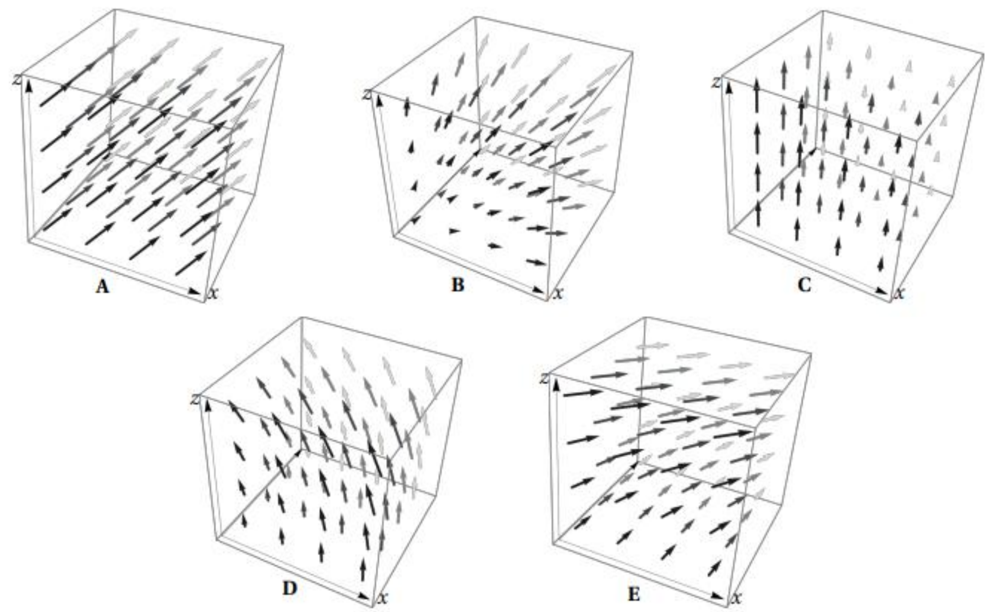
We see that $\mathbf{G} = \langle y-1, y\rangle$.

6. Here are plots of 6 vectors fields on the box where $0 \le x \le 1$, $0 \le y \le 1$, and $0 \le z \le 1$. Which of the following vector fields corresponds to the vector field $\mathbf{F} = \langle z, 1, 0\rangle$?



It should be the case that the vectors should be parallel to the *xy*-plane. The *x* component should increase as *z* values increase, and *y* values should be 1 for each vector. We can see that **D** meets this description.

7. Here are plots of 5 vectors fields on the box where $0 \le x \le 1$, $0 \le y \le 1$, and $0 \le z \le 1$. Which of the following vector fields corresponds to the vector field $\mathbf{F} = \langle -z, 0, 1+x \rangle$?



It should be the case that the vectors should be parallel to the *xz*-plane. The *x* component should decrease as *z* values increase, and *z* values should increase as *x* values increase. We can see that **D** meets this description.

8. Determine if the vector field $\mathbf{F}(x,y) = \left\langle \ln(y) + 2xy^3, 3x^2y^2 + \frac{x}{y} \right\rangle$ is conservative.

We are looking for a function $f(x,y)$ such that $\vec{\nabla} f(x,y) = \mathbf{F}(x,y)$. We can see that

$$f_x(x,y) = \ln(y) + 2xy^3 \qquad \qquad f(x,y) = x\ln(y) + x^2y^3 + c(y)$$
$$\text{which implies that}$$
$$f_y(x,y) = 3x^2y^2 + \frac{x}{y} \qquad \qquad f(x,y) = x^2y^3 + x\ln(y) + d(x)$$

Where $c(y)$ represents a function in terms of *y* and $d(x)$ represents a function in terms of *x*. We can then see that $f(x,y) = x^2y^3 + x\ln(y) + k$ (for some constant *k*).

9. Determine a potential function for the vector field $\mathbf{F}(x,y) = \langle y + 2x, x + 3y^2 + \sin(y) \rangle$.

We are looking for a function $f(x,y)$ such that $\vec{\nabla} f(x,y) = \mathbf{F}(x,y)$. We can see that

$$f_x(x,y) = y + 2x \qquad \qquad f(x,y) = xy + x^2 + c(y)$$
$$\text{which implies that}$$
$$f_y(x,y) = x + 3y^2 + \sin(y) \qquad \qquad f(x,y) = xy + y^3 - \cos(y) + d(x)$$

Where $c(y)$ represents a function in terms of *y* and $d(x)$ represents a function in terms of *x*. We can then see that $f(x,y) = xy + x^2 + y^3 - \cos(y) + k$ (for some constant *k*).